

Yin yoga as a fascia-oriented practice

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Introduction

In the modern era, authors use “bullet points” to highlight the logical structure of their writings. The bullet points of ancient India were terse statements that were easy to memorize. Each statement was called a “sutra”. In this chapter, a slightly modified version of the yin yoga sutras is presented, which we use in our training program.

All forms of yoga aim to increase flexibility and relax muscular tension. Yang yoga also emphasizes strength and endurance. When practicing yang yoga, muscles are contracted, poses are performed repetitively, and are typically held for 10 to 30 s.

Yin yoga emphasizes relaxing fascial contracture, including ligaments, joint capsules and discs. For this reason, yin poses are performed with the muscles relaxed and held for several minutes. Yin poses are primarily floor poses.

Yin yoga sutras

The purpose of yoga asana is to harmonize the flow of chi in the meridians. This is accomplished by stressing 14 skeletal segments and 10 myofascial groups.

The 14 skeletal segments are toes, ankles, knees, hips, pelvis, lumbar, thoracic, cervical, fingers, wrist, radius, ulna, humerus and scavicle. (Note: scapula + clavicle = scavicle.)

The 10 myofascial groups are groin, quads, hip flexors, glutes, hamstrings, rectus abdominis, obliques, thoracolumbar, cervical and scapular.

The two fundamental stresses are tension and compression.

Stress is created by seven asana archetypes. These archetypes are Shoelace, Saddle, Caterpillar, Dragonfly, Twist, Dog and Dragon. The seven archetypes must be adapted to the flexibility and the skeletability of every student. Flexibility refers to the extensibility of soft tissue: myofascia, joint capsules and discs. Skeletability refers to the potential range of motion of a joint segment, the range of motion before bone compression stops the movement. It is not always desirable to develop maximum skeletability. Every bone in every body is different, so what is easy for one skeleton may be impossible for another. Every yogi, therefore, has a unique skeletability. Consequently, there is no such thing as a “perfect pose”. If you are effectively stressing the target areas then you are performing the poses correctly. Every hand and foot position either helps or inhibits your ability to stress the target areas. The most effective way to do this varies from person to person.

Feel the rebound. In yin yoga, the rebound is the subjective experience of energy movement within the body. It is accompanied by pleasant feelings of physical ease, mental stillness and

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emotional calm. After practicing a pose for a few minutes, relax in any comfortable position and feel the rebound.

Skeletal joints and skeletal segments

A joint is formed when two bones are in contact. There are approximately 206 bones in the body and their various contacts create over 300 joints. But when considered as functional groups, there are only 14 joint segments, or 24 if you include both arms and both legs.

In yin yoga, the joint segments are labeled with simple mnemonic names and can include more than one joint. For example: the “lumbar segment” includes the 10 facet joints and five intervertebral discs of the five lumbar vertebrae.

When practicing yoga it is not possible to isolate the joints within a segment. For example, it is not possible to twist, bend or stretch just one lumbar vertebra. A yogi must be content to stress

all of the lumbar vertebrae. It is sometimes possible to emphasize one particular part of a segment, but all of the joints in the segment will still be affected by a yoga pose.

Muscles and myofascial groups

There are over 600 muscles in the body, but for yoga practitioners it is sufficient to consider the muscles of the body as divided into 10 myofascial groups (or 20 if you include left and right sides). Each group is formed by a shared fascial structure or fascial compartment.

The first eight myofascial groups are involved in the vast majority of yin yoga poses. These eight groups can be schematized as three myofascial groups on each side of the torso and five myofascial groups in each thigh. Each of these groups is clearly delineated by fascial compartments (Grant, 1948; Schuenke et al., 2006).

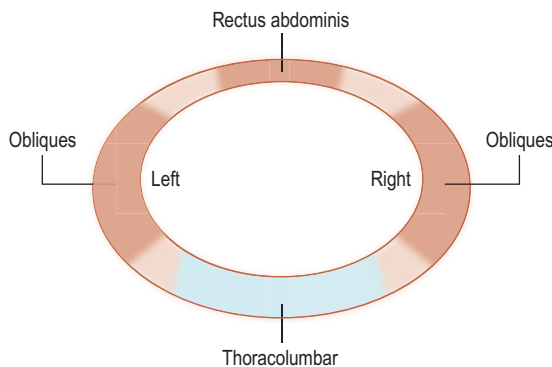


Figure 38.1

The three myofascial groups on each side of the torso can be schematically represented as above. If a yoga practitioner feels the stress of a yoga asana in the specified target area then he or she is performing the asana correctly.

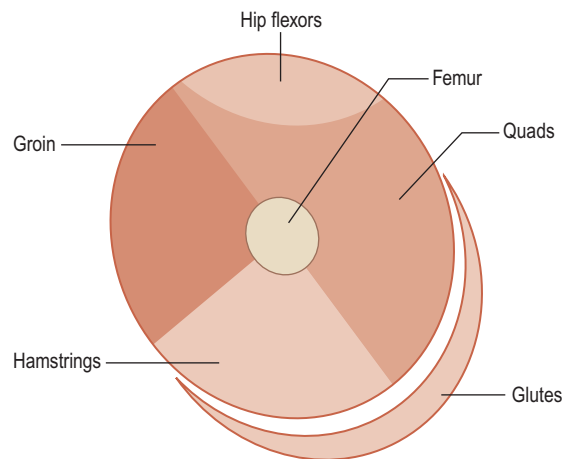


Figure 38.2

The five myofascial groups of the right thigh can be schematically represented as above. If a yoga practitioner feels the stress of a yoga asana in the specified target area then he or she is performing the asana correctly.

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Again, it is not possible to isolate just one of the muscles within a group. For example, a yogini cannot do a forward bend and stretch only one of the muscles ensheathed by the thoracolumbar fascia. A yogini is obliged to stretch all the muscles within that group. It is sometimes possible to emphasize one, but all of the muscles in the group will be affected by a yoga pose.

Yoga practice as stressing specific target areas

Presenting yoga as a practice designed to stress joint segments and myofascial groups allows us to sweep aside the various “rules of alignment” floating around in the yoga world and replace them with a system that appears to be simpler, safer and more effective. This simplification is made possible by the fascial compartmentalisation of muscles.

Six-step protocol

This is the protocol for determining how each yoga asana should be developed:

1. Select a target area to be stressed.
2. Select the asana archetype that stresses that area.
3. Select a variation of the archetype that suits your skeletability.
4. Practice the pose for 3–5 min.
5. Be sure not to overstrain the skeletal or myofascial tissues.
6. Rebound for 1 or 2 min.

The first three steps of the protocol are illustrated below.

Stressing the gluteal group: the first three steps

Step 1

Select the target area. The target area in this example will be the gluteal myofascial group, which includes 10 muscles and the iliotibial band. The muscles are gluteus minimus, gluteus medius, gluteus maximus, the six deep rotator muscles, and the tensor fasciae latae. It is not likely that all of these muscles will be affected, but all of them are potential targets. This varies from student to student.

Step 2

Select the asana archetype that stresses the gluteal group. In this instance, it is the Shoelace. All variations of the Shoelace archetype require different degrees of external rotation, abduction and flexion of the femur. The combination of these movements stresses the gluteal group.

Step 3

Select the variation of the archetype that suits your skeletability. For this demonstration both models have selected Sleeping Swan. Skeletability refers to the range of motion that is determined by the shape of your bones. In Sleeping Swan, the relevant bones are the hip socket and the femur.

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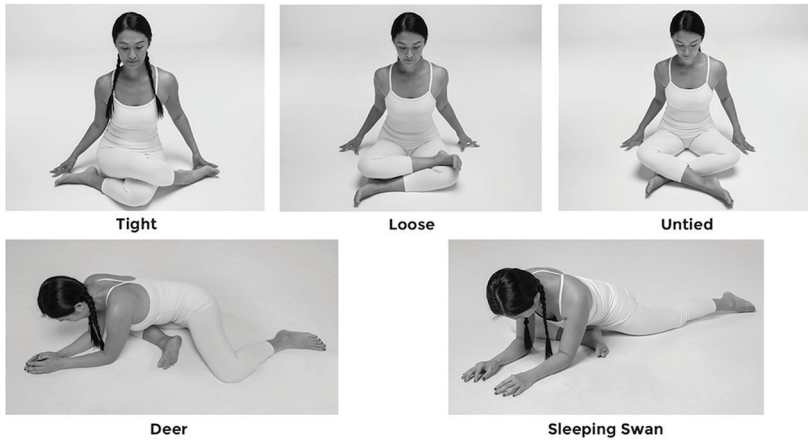


Figure 38.3

Each one of these poses is a variation of the Shoelace. Each variation externally rotates, abducts and flexes the femur. It is the combination of these movements that stresses the gluteal myofascial group. Each variation emphasizes a slightly different combination of these three movements. It is up to the student to carefully test and determine the appropriate variation for her body. For this demonstration, each model will choose the Sleeping Swan.

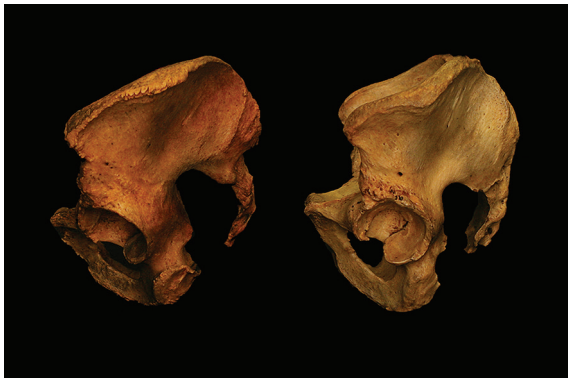


Figure 38.4

This is an image of two pelvi facing to the reader's left. The hip socket on the left pelvis is oriented forward and down. The hip socket on the right pelvis is oriented towards the camera and nearly horizontal. Without going into detail, I hope that the reader can appreciate that the amount of rotation, abduction and flexion would be dramatically different for these dramatically different hip sockets. Also, keep in mind that the differences in the femur bones are just as dramatic.



Figure 38.5

The models sitting in the chairs are allowing their left femurs to be passively rotated by a partner. Each of the models is demonstrating the limit of their ability to abduct and externally rotate their femurs. Each of them is experiencing compression of the femur against the lip of the hip socket. This type of compression is not painful, which is why they are smiling.

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Figure 38.6

The same two models are demonstrating the preferred placement of their left foot and knee. The model on the left has minimized the amount of external rotation and abduction; the model on the right has maximized them. If the model on the left pushed her foot forward and her knee out to the side then she might hurt her knee. If the model on the right brought her foot down and her knee in she would not stress the gluteal group.



Figure 38.7

Both models are now lying prone over their left leg. Both of these variations of Sleeping Swan are effectively stressing the gluteal myofascial group. There is no perfect or best variation that would be equally effective for both of them.

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Stressing the joint segments of lower body: the first three steps

Step 1

Select the target areas. The target areas in this example will be the skeletal segments of the lower body: toes, ankles, knees, hips, pelvis and



Figure 38.8

This is the Saddle archetype. The student sits on her feet, modestly abducts her femurs, and reclines. All the joints of the lower body are stressed, as well as the quad and hip flexor myofascial groups.

the lumbar spine. These joint segments are particularly susceptible to stiffness and degeneration in our modern, chair-bound society.

Step 2

Select the asana archetype that stresses these segments. In this instance it is the Saddle. Saddle also stresses two myofascial groups: the quads and hip flexors. But for this analysis, we will focus on the joint segments.

Step 3

Select the variation of the archetype that suits your skeletability. In this example, the variation is foot position. Different foot positions require different degrees of internal rotation of the hip joints and external rotation of the tibia. This analysis will focus on the internal rotation of the hip joint.

Props are frequently used in yin yoga. They are used to reduce tension in the soft tissues or to reduce compression on the bones and discs. Saddle is the epitome of a pose that frequently requires the use of props because so many joint segments are stressed.



Figure 38.9

The models sitting in the chairs are allowing their left femurs to be passively rotated by a partner. Each of the models is demonstrating the limit of their abilities to internally rotate their femurs. Each of them is experiencing compression of the femur against the lip of the hip socket. This type of compression is not painful, which is why they are smiling.

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Figure 38.10

The models are sitting on their feet. A line has been drawn to indicate the orientation of the knee. This line approximately divides the condyles of the femur; it is intended only as a reference.



Figure 38.11

In this image both models are attempting to sit between their feet. Sitting between their feet requires more internal rotation than sitting on their feet. The model on the right has very little internal rotation, as is indicated by the very slight change in the vertical lines. And so her feet have moved only slightly apart. The model on the left has a large range of internal rotation, as is indicated by the vertical line becoming nearly horizontal.

Not every student has the props available or the skill to use them as demonstrated in Figure 38.14. In this case there are two possible options:

1: Be content to sit on your feet as in Figure 38.11, or with feet apart and using a cushion as in

Figure 38.12. This benefits the toes, ankles, knees and the quads.

2: If practicing at home, kneeling on the floor and leaning back against a couch is stable and effective.

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Figure 38.12

The model on the right has minimized femoral rotation by sitting on a cushion and keeping her feet just hip-width. The model on the left has maximized femoral rotation by sitting between her feet and dorsiflexing her ankles. Students with a lot of internal rotation frequently prefer this position as it more effectively stresses the quads and hip flexors.



Figure 38.13

Both models are showing their preferred Saddle variation. The model on the right sits on a cushion to prevent knee strain. She also rests her upper body on a cushion to reduce pelvic and lumbar extension.

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Figure 38.14

Both models are showing their preferred Saddle variation. The model on the right sits on a cushion to prevent knee strain. She also rests her upper body on a cushion to reduce pelvic and lumbar extension.

Remember: it is not always desirable to develop the maximum skeletability of a joint segment.

Stressing the muscles and joints of upper body: the first three steps

Step 1

Select the target areas. The target areas in this example will be the skeletal segments and muscles of the upper body. The skeletal segments include the fingers, wrist, radius, ulna, humerus and scapula (scapula + clavicle.) The myofascial group is the scapular group. This large group includes the muscles that move and stabilize the scapula and the humerus. The scapula muscles include the rhomboids, levator scapulae, pectoralis minor, latissimus dorsi and trapezius. The humeral muscles

include the four rotator cuff muscles, deltoid, pectoralis major, biceps, coraco-brachialis, triceps and latissimus dorsi.

Step 2

Select the asana archetype that stresses these target areas. In this instance, it is the Dog. The identifying characteristic of the Dog is bearing weight on the arms.

Step 3

Select the variation of the archetype that suits your skeletability. In this example, the variation will be Chaturanga, a push-up. All the joints of the upper body will have some influence on this pose, but three are most relevant: the scapula, the humerus and the ulna. This analysis will focus on the torsion of the humerus.

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Figure 38.15

This is an image of two right humerus bones. It is the view you would have looking down your right arm. The round, proximal end of the humerus articulates with the scapula to form the shoulder joint. The distal end articulates with the ulna to form the elbow joint. The long bones of the body have torsion, they spiral, but no two long bones are identical. To demonstrate this a line has been drawn through the bicipital grooves of each specimen and these lines are oriented vertically. We could say that the elbow joint of the left specimen is "turned in" relative to the right specimen.



Figure 38.16

These two models are standing relaxed. The inner crease of their elbow joints are turned out on the left model and turned in on the right model. This is indicative of their different humeral torsions.



Figure 38.17

Flexing their elbows brings their arms and hands into very different orientations relative to their torsos. When practicing Chaturanga, the model on the left would have to internally rotate his humeri to place his palms on the floor. The model on the right would have to externally rotate his humeri to place his palms on the floor. The stresses on their joint capsules, rotator cuff and deltoid muscles are very different when they practice Chaturanga.

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Figure 38.18

A common “alignment cue” in Chaturanga is “squeeze the scapula together”. In this image, both models have retracted their scapula. The model on the left has now placed his arms in an even more disadvantageous position. He would have to internally rotate 70 degrees to place his palms on the floor. But the model on the right now has perfect arm position. He would not need to rotate his humeri at all.



Figure 38.19

Another common “alignment cue” in Chaturanga is “squeeze the elbows into the ribs”. The model on the left, because he needs to internally rotate his humeri 70 degrees, feels an uncomfortable strain in his shoulder joints if he holds his elbows to his ribs, so he holds his humeri at an angle away from his body.

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Many variations of an archetype

I do not want to leave the reader with the impression that, for each student, there is one and only one best variation of an archetype. A yogini might practice many variations of an archetype, as long as each variation effectively stresses the target areas.

In our first example, the Sleeping Swan was selected to compare two different skeletons performing the same variation. But each model might also enjoy practicing other variations of the Shoelace. Shoelace stresses the gluteal group. Each variation of Shoelace alters the abduction, rotation and flexion of the femur. These variations will almost certainly emphasize slightly different parts of this large myofascial group. It is important to explore all the variations of an archetype to determine which are effective and which are ineffective stressors of the gluteal group. Every variation that is effective should stay in your repertoire.

In our second example, we saw that the model with excellent internal rotation of the femur was comfortable sitting between her feet. But someone with this skeletability might also enjoy sitting on their feet. Sitting on their feet will elevate their pelvis. This might create a desirable extension of the lumbar spine that one does not experience when sitting between their feet. They might also experience a better stretch on their hip flexors when sitting on their feet.

In our third example, we saw that the model on the right could easily hold his elbows in to his ribs when practicing Chaturanga. But he might choose several different abduction angles of his humerus to emphasize different muscles within the scapular group. Arms abducted emphasizes

the pectoral and deltoid muscles. Arms adducted emphasizes the triceps.

Modern meridian theory

The man who initiated my interest in fascia was my teacher Dr. Hiroshi Motoyama, who had PhDs in both Philosophy and Physiological Psychology. He began his scientific investigation of meridians in the 1970s. In 1987, in an address to some of his students, he said:

“Modern Western medicine does not yet understand the meridian system. This is because the meridians lie in a water-rich phase within the connective tissue (fascia). When a person dies, this layer dries out. Since the meridian channels have no walls as the blood vessels do, they cannot be detected by current medical techniques. Actually, however, these water-rich channels crisscross the entire body.”

Motoyama (2003).

It is no longer true that practitioners of Western medicine do not understand the meridian system. Indeed, many sides of scientific inquiry appear to be closing in on the meridian system. Two findings of modern fascia research are most relevant to yoga practitioners: fascia’s ability to contract and relax, and fascia’s ability to absorb, bind and release water.

Rebound and rehydration

It has been demonstrated that slow, compressive massage strokes temporarily force water out of myofascial tissues, and that when the tissues recover the water content is higher than before. It has also been demonstrated that relaxing after long, static stretching can induce a higher hydration in fascia (Schleip et al., 2012).

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This super-compensation of the water content aptly describes the subjective experience of the rebound after a yoga pose.

After practicing a pose for several minutes, a yin yogini will rest for a minute or two and focus on what she is feeling in the target area. This period of restful introspection is called “the rebound”. During the rebound a yogini typically feels fluid movement and a sense of heat dispersion or vibration. It is innately pleasant and relaxing.

Meridians conduct energy

Dr. Motoyama demonstrated that the water-rich channels in fascia are electrically conductive. He also demonstrated that the currents in these channels propagate at a different velocity than the nervous system, that the pathways of conduction are different from those of the nervous system, and that the local response to electrical stimulus of a meridian is opposite to that of the nervous system (positive vs. negative). He also demonstrated that the pathways of the watery fascial channels correspond very closely to the pathways described in ancient acupuncture texts (Motoyama, 1997).

If and when these findings are confirmed by other researchers, this would have large implications for acupuncture theory, yoga theory and modern medicine. If experiments were to confirm that the interstitial environment has malleable qualities of electrical capacitance and conduction, then the colloquial “I am having a low energy day” would take on a medically testable meaning.

Motoyama noted that a certain electrical response, which he called the “before polarization” current, was higher in healthy people

who had good fluid flow in their fascia, and lower in elderly, seriously ill or physically weak people, who had poor fluid flow in their fascia (Motoyama, 1997).

One day, I hope that the following statement is a scientifically validated fact:

Some of the mechanical stresses created by yoga postures are converted into greater electrical capacitance and conductance in the meridians. This affects the migration of various cells through the fascial/interstitium, the movement of substances in and out of cells, and alters what cells express. This is why the healing effects of the practice go far beyond flexibility and relaxation.

The rebound is more than physical

During the rebound, yogis also experience emotional contentment and mental calm. A simple theory of fascial rehydration cannot explain the depth of the emotional and mental responses. A yogini would say they are the result of harmonizing the flow of her chi.

Chi is the ancient Chinese term for life force. The theory of a life force was hotly debated and dismissed from modern medicine over 100 years ago but the conception of a life force is central to acupuncture theory and yoga practice. Dr. Motoyama taught that chi is a unique form of energy not reducible to electricity or chemistry.

The ancient theory of chi and the modern theory of meridians are an accurate description of the subjective experience of yoga. I believe there are watery meridian channels penetrating and sustaining all tissues of the body, including bones, joints, viscera and the cerebrospinal fluid. The purpose of yoga asana is to harmonize the flow of chi in these meridians.

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Other resources

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On the web: CIHS.edu
701 Garden View Court
Encinitas, CA 92024.

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Wild Strawberry Productions

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Youtube: Piriformis Muscle Medially Rotates with Dr. Joe Muscolino.

<https://www.youtube.com/watch?v=vGZPw6ARj0s>

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